Artificial Intelligence Midterm Study Guide

CSE 316 / CSE 411 | Daffodil International University

সহজ ইংরেজি + বাংলা | Beginner Level

Based on your uploaded files:
- Topic 1: Introduction to AI (lecture slides)
- Midterms: Spring 2026, Fall 2025, Summer 2025, Spring 2025

Contents

Part 1 – Lecture in easy English + বাংলা
Part 2 – 3 past questions + 3 practice questions (Scenario → Question → Answer)
Part 3 – Colour-coded probability table + 5 exam tips

Note: The lecture slides cover only Introduction to AI. Search, genetic algorithm and expert-system answers use standard AI knowledge, because those topics are in the exams but not in the slides.

PART 1: Lecture 1 – Introduction to AI (এআই-এর ভূমিকা)

Step 1. What is AI? (এআই কী?)

English: AI (Artificial Intelligence) is a branch of computer science. It builds computer systems that show some form of intelligence. These systems can:

- learn new concepts and tasks,
- reason and draw useful conclusions about the world,
- understand human language, or see and understand a visual scene,
- do other tasks that normally need human intelligence.

বাংলা: এআই হলো কম্পিউটার সায়েন্সের একটি শাখা। এখানে কম্পিউটারকে এমনভাবে তৈরি করা হয় যেন সে নতুন জিনিস শেখে, চিন্তা করে সিদ্ধান্ত নেয়, মানুষের ভাষা বোঝে, ছবি দেখে বোঝে এবং মানুষের বুদ্ধি লাগে এমন কাজ করতে পারে।

Four ways to define AI (এআই-কে ৪ ভাবে সংজ্ঞায়িত করা যায়)

	Think (চিন্তা)	Act (আচরণ)
Like humans (মানুষের মতো)	Systems that think like humans	Systems that act like humans
Rationally (যুক্তিসংগতভাবে)	Systems that think rationally	Systems that act rationally

সহজ কথা: মানুষের মতো আচরণ করা, নাকি সবসময় সবচেয়ে ভালো সিদ্ধান্ত নেওয়া। শেষেরটিকে (act rationally) *rational agent* বলে। এটি এই কোর্সের মূল ধারণা।

Step 2. Major Branches of AI (এআই-এর প্রধান শাখা)

Branch	Easy meaning (সহজ অর্থ)
Machine Learning, Artificial Neural Networks	ডেটা থেকে শেখা; মস্তিষ্কের নিউরনের মতো মডেল
Computer Vision, Pattern Recognition	ছবি ও প্যাটার্ন চেনা
Natural Language Processing	মানুষের ভাষা বোঝা ও বলা (যেমন চ্যাটবট)
Expert Systems, Knowledge Management	বিশেষজ্ঞের জ্ঞান নিয়ম আকারে রেখে সিদ্ধান্ত দেওয়া
Fuzzy Systems	"একটু গরম", "বেশ ঠান্ডা" ধরনের অস্পষ্ট তথ্য সামলানো
Heuristic Search, Metaheuristic & Swarm Intelligence	সমাধান খোঁজার স্মার্ট পদ্ধতি (A*, জেনেটিক অ্যালগরিদম, পিঁপড়ার দল)
Game AI	খেলায় চাল দেওয়া (টিক-ট্যাক-টো, দাবা)
Robotics, Virtual Intelligence	রোবট; ভার্চুয়াল পরিবেশে বুদ্ধিমান আচরণ

Step 3. Foundations of AI (এআই যে বিষয়গুলো থেকে এসেছে)

Discipline	Since	Main question	বাংলা
Philosophy	428 B.C.	Can formal rules give valid conclusions? Where does knowledge come from?	নিয়ম দিয়ে সঠিক সিদ্ধান্ত নেওয়া যায় কি? জ্ঞান কোথা থেকে আসে?
Mathematics	800 B.C.	What can be computed? How to reason with uncertainty? (algorithms, NP-completeness, probability)	কী কম্পিউট করা যায়? অনিশ্চিত তথ্যে যুক্তি কীভাবে?
Economics	1776	How to decide to maximize payoff? (decision theory = probability + utility)	লাভ সর্বোচ্চ করে সিদ্ধান্ত কীভাবে?
Neuroscience	1861	How does the brain process information? (neurons)	মস্তিষ্ক কীভাবে তথ্য প্রসেস করে?
Psychology	1879	How do humans and animals think and act?	মানুষ ও প্রাণী কীভাবে ভাবে ও কাজ করে?
Computer Engineering	1940	How can we build an efficient computer?	কার্যকর কম্পিউটার কীভাবে বানাব?
Cybernetics	1948	How can artifacts operate under their own control?	যন্ত্র নিজের নিয়ন্ত্রণে কীভাবে চলে?
Linguistics	1957	How do languages relate to thought?	ভাষা ও চিন্তার সম্পর্ক কী?

Cybernetics vs AI. The word *cybernetics* was coined by **Norbert Wiener**, an American mathematician. It is the study of communication and control in biological, mechanical and electronic systems. The slides say AI and Cybernetics are **not** the same subject, though many people think so.

> **বাংলা:** *Cybernetics* শব্দটি দিয়েছেন আমেরিকান গণিতবিদ **নরবার্ট উইনার**। এটি জীব, যান্ত্রিক ও ইলেকট্রনিক সিস্টেমে যোগাযোগ ও নিয়ন্ত্রণ নিয়ে পড়াশোনা। এআই আর সাইবারনেটিক্স এক বিষয় নয়।

Step 4. History of AI (এআই-এর ইতিহাস)

Period	Name	Simple meaning (সহজ অর্থ)
1943–1955	Gestation of AI	ধারণার শুরু (ভ্রূণ অবস্থা)
1956	Birth of AI	"এআই" নামটির জন্ম
1952–1969	Early enthusiasm, great expectations	প্রথম সাফল্যে অনেক আশা
1966–1973	A dose of reality	বাস্তবতা বোঝা (স্লাইডে এখানে Genetic Algorithm লেখা)
1969–1979	Knowledge-based systems	বিশেষজ্ঞ জ্ঞানভিত্তিক সিস্টেম
1980–present	AI becomes an industry	এআই ব্যবসায় পরিণত
1986–present	Return of neural networks	নিউরাল নেটওয়ার্ক ফিরে এল
1987–present	AI becomes a science	এআই বৈজ্ঞানিক পদ্ধতিতে এল

Step 5. State of the Art (এখনকার ব্যবহার)

Autonomous planning and scheduling • game playing • autonomous control • medical diagnosis • logistics planning • robotics • language understanding and problem solving.

> স্বয়ংক্রিয় পরিকল্পনা, খেলা, নিয়ন্ত্রণ, রোগ নির্ণয়, লজিস্টিকস, রোবট, ভাষা বোঝা ও সমস্যা সমাধান।

Step 6. Philosophical Foundations (দার্শনিক ভিত্তি)

Philosophers ask three questions: (1) How do minds work? (2) Can machines act intelligently like people, and would they have real conscious minds? (3) What are the ethical implications of intelligent machines?

Term	Meaning	বাংলা
Weak AI	Machines can *act as if* they are intelligent.	যন্ত্র বুদ্ধিমানের মতো **আচরণ** করতে পারে।
Strong AI	Machines that do so are *actually thinking*, not just simulating thinking.	যন্ত্র সত্যিই **চিন্তা** করে।

The Turing Test (টুরিং টেস্ট)

Alan Turing said: instead of asking "Can machines think?", ask "Can a machine pass a behavioural intelligence test?"

1. A program has a conversation with an interrogator by **typed messages for 5 minutes**.
2. The interrogator guesses: is it a human or a program?
3. The program **passes** if it fools the interrogator **30%** of the time.

Facts to remember:

- Turing predicted a computer with 10^9 units of storage would pass the test by the **year 2000**. He was **wrong**: programs could not fool a sophisticated judge.
- People were fooled when they did not know it was a computer: **ELIZA, MGONZ, NATACHATA, CYBERLOVER**.
- **Eugene Goostman**: chatbot made in Saint Petersburg in **2001** by Vladimir Veselov, Eugene Demchenko and Sergey Ulasen. It finished 2nd in the Loebner Prize in 2005 and 2008.
- June **2012**: it convinced **29%** of judges. 7 June **2014**: **33%** of judges thought it was human; organiser **Kevin Warwick** said it passed. Critics questioned this.
- Few AI researchers care about the Turing test; they focus on practical performance. Arguments for and against strong AI are inconclusive.

বাংলা: টুরিং টেস্ট হলো ৫ মিনিটের টাইপ করা কথোপকথন। প্রোগ্রাম ৩০% বিচারককে বোকা বানাতে পারলে পাস। টুরিং ভেবেছিলেন ২০০০ সালের মধ্যে পাস হবে, কিন্তু তিনি ভুল ছিলেন। Goostman ২০১৪ সালে ৩৩% পেয়েছিল, কিন্তু সমালোচকরা দাবিটি মানেননি।

Step 7. Logic Programming and Prolog (লজিক প্রোগ্রামিং)

- In logic programming, a program is a set of **facts and rules**. An **inference engine** answers queries by searching them.
- Main families: **Prolog, Answer Set Programming (ASP), Datalog**.
- **Prolog = Programming in Logic.** Queries are answered by a method called **resolution**.

```
parent(ann, sam).     % fact: ann is a parent of sam
parent(joe, ann).     % fact: joe is a parent of ann
male(joe).            % fact
grandfather(X, Z) :- parent(X,Y), parent(Y,Z), male(X).   % rule

?- grandfather(joe, sam).     % query  -->  true
```

বাংলা: `:-` মানে "যদি" (if) এবং `,` মানে "এবং" (and)। joe হলো ann-এর parent, ann হলো sam-এর parent, আর joe পুরুষ। তাই joe হলো sam-এর grandfather, উত্তর **true**।

PART 2: Past Questions and Practice Questions (পূর্বের প্রশ্ন ও অনুশীলনী)

PAST = from your real exam papers. **PRACTICE** = for extra practice. Each has **Scenario → Question → Answer (step by step)**.

PAST 1 Intelligent Agent (Spring 2026, Q1a, 5 marks)

SCENARIO

The DIU playground is covered with green grass that grows very fast and must be cut every week. Manual work is very difficult. The CSE department asks you to design an autonomous agent to solve the problem.

QUESTION

Define intelligent agent. For the scenario, describe the strategy, planning and type of the designed agent. Justify your answer.

ANSWER

Step 1 **Define the agent.** An intelligent agent is anything that **perceives** its environment through **sensors** and **acts** on it through **actuators** to reach its goal in the best possible way. (যে সেন্সর দিয়ে পরিবেশ দেখে এবং অ্যাকচুয়েটর দিয়ে কাজ করে লক্ষ্য অর্জন করে।)

Step 2 **Write the PEAS.**

Performance	% of grass cut evenly, time taken, battery used, safety
Environment	Playground, grass, obstacles (goalposts, stones), people, rain
Actuators	Wheels, cutting blades, motors
Sensors	Camera, GPS, bump/obstacle sensor, grass-height sensor, rain sensor, battery sensor

Step 3 **Strategy.** Use **coverage path planning**: move in a back-and-forth (zig-zag) pattern over the whole field, so no area is missed or repeated. Avoid obstacles and stop if a person comes close.

Step 4 **Planning.** (1) Map the field. (2) Plan a route that covers every cell. (3) Schedule one cut every week. (4) Go back to the charger when the battery is low, then continue. (5) Re-plan when a new obstacle appears.

Step 5 **Type of agent: utility-based (goal-based) agent.** The goal is "all grass cut". It also compares options (shortest route vs battery left vs safety), so a utility function makes it better than a simple goal-based agent.

Step 6 **Task environment.** Partially observable, stochastic (people and weather change), dynamic, sequential, continuous.

> **বাংলা সারাংশ:** রোবট সেন্সর দিয়ে মাঠ দেখবে, জিগ-জ্যাগ পথে ঘাস কাটবে, বাধা এড়াবে, ব্যাটারি কমলে চার্জে ফিরবে। এটি utility-based agent, কারণ সে একাধিক বিকল্পের মধ্যে সবচেয়ে ভালোটি বেছে নেয়।

PAST 2 A* and Uniform Cost Search (Summer 2025, Q2d, 4 marks)

SCENARIO

A directed graph has start **S** and goal **G**.
Edge costs: S→A 6, S→B 3, A→D 5, A→E 12, B→E 10, B→C 7, C→E 2, D→G 16, E→G 7.
Heuristic h(n): S=14, A=12, B=11, C=6, D=11, E=4, G=0.

QUESTION

Find the optimum path from S to G using A*. Is it possible to apply Uniform Cost Search (UCS)? Compare the results.

ANSWER

Step 1 **Formula.** A* uses **f(n) = g(n) + h(n)**. g = cost from start so far; h = estimated cost to goal. Always expand the node with the smallest f.

Step 2 **Run A*.**

#	Expand	New nodes in frontier (g, f)
1	S (f=14)	A (g=6, f=18), B (g=3, f=14)
2	B (f=14)	E (g=13, f=17), C (g=10, f=16)
3	C (f=16)	E improved through C: g=12, **f=16**
4	E (f=16)	G (g=19, **f=19**)
5	A (f=18)	D (g=11, f=22)
6	G (f=19)	**Goal reached**

Step 3 **Result.** Path **S → B → C → E → G**, cost = 3 + 7 + 2 + 7 = **19**. Expansion order: S, B, C, E, A, G (6 nodes).

Step 4 **UCS.** Yes, UCS can be applied, because all edge costs are positive. UCS uses only g(n) and no heuristic. Order: S(0), B(3), A(6), C(10), D(11), E(12), G(19), which is 7 nodes. It gives the **same path and cost 19**.

Step 5 **Compare.** Both are optimal. A* expanded fewer nodes (6 vs 7) because h(n) pushes it toward the goal. UCS searches in all directions by cost only.

> **বাংলা:** A* ও UCS দুটোই সবচেয়ে সস্তা পথ S-B-C-E-G (খরচ ১৯) পায়। A* কম নোড দেখে, কারণ সে heuristic ব্যবহার করে। UCS শুধু এ পর্যন্ত খরচ (g) দেখে।

PAST 3 Compare Search Strategies (Fall 2025, Q2a and Q2b, 6 marks)

SCENARIO

A delivery robot must go from warehouse **A** to delivery point **G**.
Roads (km): A→B 2, A→C 4, A→D 3, B→E 5, C→D 1, C→F 3, D→F 2, E→G 3, F→G 6.
h(n): A=7, B=5, C=1, D=3, E=2, F=2, G=0.

QUESTION

(a) List and compare the order in which nodes are expanded for UCS, Greedy best-first, A* and Iterative Deepening Search. (b) Which algorithm found the goal faster (fewer nodes)? Is the heuristic admissible and consistent? Justify.

ANSWER

Step 1 **UCS** (by g only): A(0), B(2), D(3), C(4), F(5), E(7), G(10). Path A-B-E-G, cost **10**. 7 nodes.

Step 2 **Greedy** (by h only): A → C (h=1) → F (h=2) → G. Order A, C, F, G. Path A-C-F-G, cost **13**. 4 nodes.

Step 3 **A*** (f = g + h): A(7), C(5), D(6), then B and F tie at f=7, then E(9), G(10). Path A-B-E-G, cost **10**. 7 nodes (the tie can be broken either way; the count stays 7).

Step 4 **IDS** (repeat depth-limited DFS, limits 0, 1, 2, 3). The goal is first reachable at depth 3.
Limit 0: A | Limit 1: A, B, C, D | Limit 2: A, B, E, C, D, F, D, F | Limit 3: A, B, E, **G**. Path A-B-E-G.

Algorithm	Nodes expanded	Path	Cost	Optimal?
UCS	7	A-B-E-G	10	Yes
Greedy	4	A-C-F-G	13	No
A*	7	A-B-E-G	10	Yes
IDS	repeats earlier levels	A-B-E-G	10	Fewest steps, not always cheapest

Step 5 **(b) Faster. Greedy** expanded the fewest nodes (4) but its path is not optimal (13 vs 10). A* and UCS both gave the best path here with 7 nodes.

Step 6 **Admissible?** Yes. h(n) never overestimates the real cost to the goal. Real costs: A=10, B=8, C=9, D=8, E=3, F=6. Check: 7≤10, 5≤8, 1≤9, 3≤8, 2≤3, 2≤6.

Step 7 **Consistent? No.** Rule: h(n) ≤ cost(n, n') + h(n'). For A→C: h(A)=7 but 4 + h(C) = 5, and 7 > 5. A→D also fails: 3 + 3 = 6 < 7. So the heuristic is **admissible but not consistent**.

> **বাংলা:** Greedy সবচেয়ে কম নোড (৪টি) দেখেছে, কিন্তু সেরা পথ পায়নি। Heuristic admissible (কখনো বেশি অনুমান করেনি), কিন্তু consistent নয়, কারণ A→C ধাপে শর্ত ভাঙে।

PRACTICE 1 Genetic Algorithm (based on Fall 2025, Q2c, 4 marks)

SCENARIO

A population has four chromosomes: C1 = 101010, C2 = 110011, C3 = 100110, C4 = 111000. Use **fitness = number of 1s** in the chromosome.

QUESTION

Perform the three major operations of the genetic algorithm and show if the population improves.

ANSWER

Step 1 **Fitness.** C1=3, C2=4, C3=3, C4=3. Total = 13.

Step 2 **Selection** (fitter chromosomes get more chances): choose C2, C1, C2, C4 as parents. C3 is dropped.

Step 3 **Crossover** (single point after bit 3):
C2 `110|011` × C1 `101|010` → `110010` and `101011`
C2 `110|011` × C4 `111|000` → `110000` and `111011`

Step 4 **Mutation** (flip one bit): flip the last bit of `110000` → `110001`.

Step 5 **New population:** 110010 (3), 101011 (4), 110001 (3), 111011 (5). Total fitness = **15**, up from 13. The population improved. Repeat until fitness stops improving.

> **বাংলা:** Selection = ভালোগুলো বাছা; Crossover = দুই অভিভাবকের বিট মেশানো; Mutation = একটি বিট উল্টানো। ফলে মোট fitness ১৩ থেকে ১৫ হলো।

PRACTICE 2 BFS, Depth-Limited and Iterative Deepening (based on Summer 2025, Q2a and Q2b, 4 marks)

SCENARIO

The start state is number 1. Each state k has two successors: 2k and 2k+1. The goal state is **11**.

QUESTION

Construct the state space for states 1 to 15. List the order in which nodes are visited for BFS, depth-limited search (limit 3) and iterative deepening search.

ANSWER

Step 1 **State space (tree).** Depth 0: 1 | Depth 1: 2, 3 | Depth 2: 4, 5, 6, 7 | Depth 3: 8, 9, 10, 11, 12, 13, 14, 15. (1 → 2,3; 2 → 4,5; 3 → 6,7; 4 → 8,9; 5 → 10,11; 6 → 12,13; 7 → 14,15.)

Step 2 **BFS** (level by level): 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, **11**.

Step 3 **DLS, limit 3** (go deep first, left child first): 1, 2, 4, 8, 9, 5, 10, **11**.

Step 4 **IDS** (run DLS with limit 0, 1, 2, 3):
Limit 0: 1
Limit 1: 1, 2, 3
Limit 2: 1, 2, 4, 5, 3, 6, 7
Limit 3: 1, 2, 4, 8, 9, 5, 10, **11** (goal found)

> **বাংলা:** BFS স্তর ধরে এগোয়, DLS আগে গভীরে যায়, আর IDS সীমা ০, ১, ২, ৩ করে বারবার DLS চালায়। শেষ সীমায় (৩) লক্ষ্য ১১ পাওয়া যায়।

PRACTICE 3 Expert System (based on Fall 2025, Q3a, 5 marks)

SCENARIO

You are designing an AI medical-diagnosis expert system for rural health centres in Bangladesh. It helps doctors identify typhoid, dengue, influenza and malaria using symptoms, test results and environment data (temperature, rainfall).

QUESTION

Illustrate the expert system: describe the knowledge base structure with at least two sample rules, the type of inference engine, and how patient data and medical records are updated over time.

ANSWER

Step 1 **Components.**

Component	Job
Knowledge Base	Doctors' knowledge stored as IF-THEN rules and facts about diseases
Inference Engine	The "brain": applies rules to facts to reach a diagnosis
Working Memory	The current patient's symptoms and test results
User Interface	Health worker enters data and reads the result
Explanation Facility	Shows *why* the system gave this diagnosis
Knowledge Acquisition	Experts add or edit rules

Step 2 **Sample rules.**
Rule 1: IF fever lasts 3 days or more AND fever is rising step by step AND abdominal pain AND Widal test positive THEN diagnosis = **Typhoid** (confidence 0.8).
Rule 2: IF high fever AND severe headache or joint pain AND low platelet count AND heavy-rainfall season THEN diagnosis = **Dengue** (confidence 0.8).

Step 3 **Inference engine type. Forward chaining** (data-driven): start from symptoms and move toward a disease. **Backward chaining** can confirm a guess: "Is it dengue? Check the platelet count."

Step 4 **Updating records.** Each visit adds new symptoms and test results to the patient's record. The engine re-runs the rules on the updated data, so the diagnosis improves over time. Experts also update rules as new cases arrive.

> **বাংলা:** Expert system-এ আছে Knowledge Base (নিয়ম), Inference Engine (সিদ্ধান্ত নেয়), Working Memory (রোগীর তথ্য) ও User Interface। লক্ষণ থেকে রোগে পৌঁছালে সেটি forward chaining।

PART 3: Probability Table and Exam Tips (সম্ভাবনা ছক ও পরীক্ষার টিপস)

These probabilities are MY ESTIMATE from the 4 past papers you uploaded. They are NOT a guarantee. (এগুলো আমার অনুমান, নিশ্চয়তা নয়।)

Step 1. Colour-coded probability table

Chance	Topic	Seen in	Why
VERY HIGH	Search algorithms (BFS, DFS, DLS, IDS, UCS, Greedy, A*)	Fall 25, Summer 25, Spring 25, Spring 26	Every paper has it; marks are high
VERY HIGH	Intelligent agent, PEAS, environment types	All 4 papers	Question 1 in every paper
HIGH	Problem formulation (goal vs problem, 8-puzzle, state space)	Spring 25, Summer 25, Fall 25	Appears in 3 of 4
HIGH	Expert system (components, rules, inference)	Fall 25, Summer 25, Spring 26	Appears in 3 of 4
HIGH	Genetic algorithm (selection, crossover, mutation)	Fall 25, Summer 25	Two papers; easy marks
HIGH	Heuristic h(n), admissible vs consistent	Fall 25, Spring 26	Linked to A*
MEDIUM	Hill climbing and local optima	Summer 25	One paper
MEDIUM	Beam search vs A*	Spring 25	One paper
MEDIUM	Lecture 1: Turing test, weak/strong AI, foundations, history, Prolog	None of the 4 papers	Not seen, but it is the base topic, so a theory question is possible

লাল = খুব বেশি সম্ভাবনা, কমলা = বেশি, হলুদ = মাঝারি।

Step 2. Five exam tips (পরীক্ষার ৫টি টিপস)

#	Tip (English)	টিপস (বাংলা)
1	**Show every step in search questions.** Draw the tree, write g, h, f values and the expansion order. Marks come from steps, not only the final path.	**সার্চের প্রশ্নে প্রতিটি ধাপ দেখান।** গাছ আঁকুন, g, h, f লিখুন। শুধু শেষ উত্তরে পুরো নম্বর নেই।
2	**Learn PEAS and the 5 environment properties by heart.** Fully/partially observable, deterministic/stochastic, episodic/sequential, static/dynamic, discrete/continuous. They appear in every paper.	**PEAS ও পরিবেশের ৫টি বৈশিষ্ট্য মুখস্থ করুন।** প্রতি পরীক্ষায় আসে।
3	**Read the action word.** *Define* = short meaning. *Explain* = give reasons. *Develop/Construct* = build with steps. *Illustrate* = diagram or example. *Compare* = use a table.	**প্রশ্নের ক্রিয়াপদ দেখুন।** Define, Explain, Illustrate, Compare প্রতিটি আলাদা ধরনের উত্তর চায়।
4	**Answer parts in order.** The paper says all portions must be answered sequentially. Label each part: a), b).	**উপপ্রশ্নগুলো ক্রমানুসারে লিখুন** এবং প্রতিটির আলাদা লেবেল দিন।
5	**Manage time.** 25 marks in 1.5 hours is about 3.5 minutes per mark. Do sure-mark parts first (agent, GA, expert system). In A* ties, write how you broke the tie.	**সময় ভাগ করুন।** ২৫ নম্বরে ১.৫ ঘণ্টা, প্রতি নম্বরে প্রায় ৩.৫ মিনিট। নিশ্চিত নম্বরের অংশ আগে করুন। টাই হলে কীভাবে ভাঙলেন লিখুন।